Solving exponential equations using logs

Because logarithms are the inverse of exponential functions, you can use logs to undo exponential equations.

$$e^x = a$$

$$\ln a = x$$

$$e^{\ln a} = a$$

Don’t confuse transforming log functions with using logs to undo exponential functions. The variable is not an exponent here.

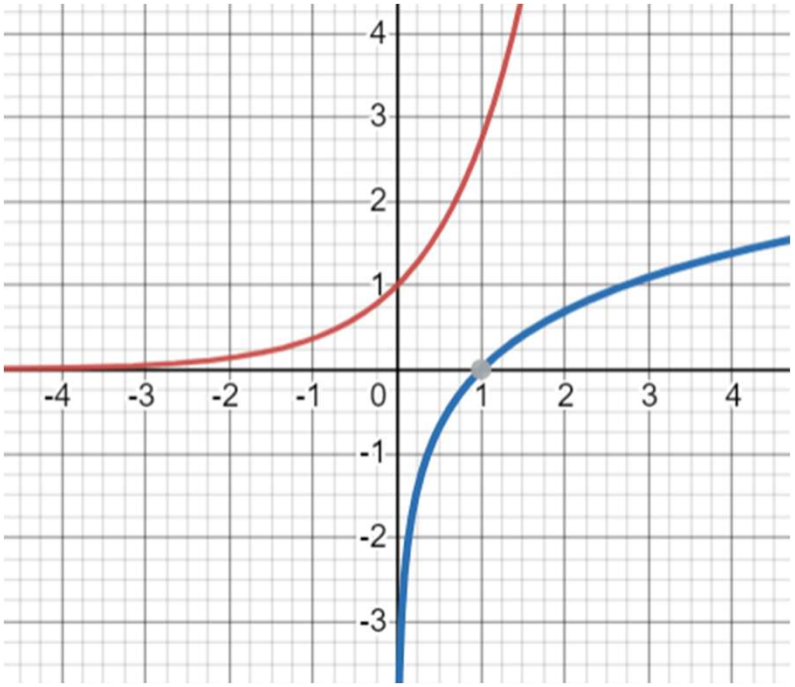
$$a \ln(x+b) + c$$

Here the variable is an exponent. Remember that the x in ln x is really an answer, not an exponent.

$$e^{\ln x} = x$$

$$\ln e^{x} = x$$

The input for the exponential function is an exponent and the output for the natural log is the exponent.



Operations on exponents are different so operations on logs are also.

$$m = e^3 \qquad n = e^2$$

$$m \cdot n = e^3 \cdot e^2 = e^{3+2}$$

$$\ln(m \cdot n) = \ln m + \ln n$$

$$5 = 3 + 2$$

When you multiply the base, you add the exponents.

So, when you multiply the input for the logarithms, you add the outputs which are exponents.

$$n = e^2 \quad m = e^3 \qquad n \cdot m = e^2 \cdot e^3 = e^{2+3}$$

$$\ln(n \cdot m) = \ln n + \ln m$$

$$\ln(e^2 \cdot e^3) = \ln e^2 + \ln e^3$$

$$5 = 2 + 3$$

$$f(x) = e^x$$

$$f^{-1}(x) = \ln x$$

Operations on exponents are different so operations on logs are also.

When you divide the bases, you subtract the exponents.

So, when you divide the input for the logarithms, you subtract the outputs which are exponents.

$$f(x) = e^x$$

$$f^{-1}(x) = \ln x$$

$$m = e^3 \quad n = e^2 \qquad \frac{m}{n} = \frac{e^3}{e^2} = e^{3-2}$$

$$\ln m - \ln n = \ln\left(\frac{m}{n}\right)$$

$$\ln e^3 - \ln e^2 = \ln\left(e^{3-2}\right)$$

$$3 - 2 = 1$$

Operations on exponents are different so operations on logs are also.

When you raise the base to a power, you multiply the exponents.

So, a base to an exponent times a constant is the same as multiplying the constant times the log

$$f(x) = e^x$$

$$f^{-1}(x) = \ln x$$

$$\ln e^p = p$$

$$(e^x)^p = a^p$$

$$\ln(e^x)^p = \ln a^p$$

$$p\ln(e^x) = p\ln a$$

To use logs to solve exponential equations, you take the log of both sides of the equation and then use the special log rules to solve.

$$e^{x+2} = 5$$

$$\ln e^{x+2} = \ln 5$$

$$x+2 = \ln 5$$

$$x = \ln 5 - 2$$

$$e^{2x} = 3$$

$$\ln e^{2x} = \ln 3$$

$$2x \ln e = \ln 3$$

$$2x = \ln 3$$

$$x = \frac{\ln 3}{2}$$

$$e^{x^2} = 4$$

$$\ln e^{x^2} = \ln 4$$

$$x^2 = \ln 4$$

$$x = \pm\sqrt{\ln 4}$$

Some calculators use ln to calculate square roots. It is easy to program a computer to calculate e to a number and to calculate the ln of a number.

So, putting the square root of a number in terms of e and ln, makes it easier to do the calculations.

$$\sqrt{a} = e^{\frac{1}{2}\ln a}$$

$$a^{\frac{1}{2}} = e^{\frac{1}{2}\ln a}$$

$$\ln a^{\frac{1}{2}} = \ln e^{\frac{1}{2}\ln a}$$

$$\ln a^{\frac{1}{2}} = \frac{1}{2}\ln a(\ln e)$$

$$\ln a^{\frac{1}{2}} = \ln a^{\frac{1}{2}} \cdot 1$$